The Neurotech EEG Dataset: A Large Clinical Scalp EEG Corpus for AI Development and Research

Authors: Keith Morgan[1], Charles Pickering[1], Matthew Goodwin[1], Han Wu[2], Manohar Ghanta[2], Aditya Gupta[2], Jin Jing[2], ChenXi Sun[2], Daniel Goldenholz[2], M. Brandon Westover[2]

[1] Neurotech, [City, State]

[2] Department of Neurology, Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA

Corresponding author: M. Brandon Westover, MD, PhD (bwestove@bidmc.harvard.edu)

Abstract

We present the Neurotech EEG Dataset, a large clinical scalp electroencephalography (EEG) corpus comprising 8,410 EEG recordings from 1,744 patients collected at a single center between 2021 and 2025, totaling 77,575 hours of signal data and 3.95 TB of storage. All recordings used Natus/Xltek hardware at 256 Hz with the standard International 10-20 montage, spanning the full spectrum of clinical EEG practice: routine outpatient recordings (34%), ambulatory and short-term monitoring (55%), and prolonged continuous monitoring (11%). Recordings are accompanied by 83,714 technician-placed annotations including 19,705 spike markers, 2,757 seizure markers, 8,027 sharp wave annotations, and free-text clinical observations. We release the dataset in Brain Imaging Data Structure (BIDS) EEG format through the Brain Data Science Platform (BDSP) with de-identification compliant with the Health Insurance Portability and Accountability Act (HIPAA), including per-patient date shifting and automated name scrubbing of free-text annotations. By preserving workflow-native annotations, this resource enables researchers to train and validate EEG algorithms against the variability and noise of real clinical practice -- bridging the gap between benchmark performance and real-world deployment.

Background & Summary

Expert interpretation of the electroencephalogram (EEG) remains the cornerstone of epilepsy diagnosis, yet the global shortage of trained EEG readers creates a bottleneck affecting the approximately 50 million people living with epilepsy worldwide[1]. Machine learning (ML) offers a path toward scalable automated interpretation[2], but the scarcity of large, clinically representative public datasets has constrained progress. Spike detection algorithms trained on existing public datasets achieve sensitivities of 70-85% on held-out test sets but can drop substantially when deployed on recordings from different clinical settings or hardware platforms -- a persistent and well-documented generalization problem[3-6].

Existing public EEG resources span a range of sizes and designs but share common limitations (Table 2). The CHB-MIT dataset provides 23 pediatric patients with seizure annotations[3]; the Bonn dataset offers intracranial recordings from 5 patients[4]; the Siena dataset contributes 14 patients with scalp EEG[5]. The Temple University Hospital (TUH) EEG Corpus, at over 25,000 sessions, demonstrated that large-scale release of unselected clinical data could become the most widely used benchmark in the EEG artificial intelligence (AI) literature[7]. However, publicly available clinical EEG remains insufficiently diverse across institutions, hardware platforms, clinical settings, and annotation practices.

Here we release the Neurotech EEG Dataset -- to our knowledge, one of the largest single-center public clinical EEG corpora -- comprising 8,410 EEG recordings from 1,744 patients totaling 77,575 recording hours. This dataset complements TUH by providing: (1) a majority of ambulatory and multi-day recordings (55% of recordings, compared with a predominantly inpatient corpus in TUH), (2) Natus/Xltek hardware enabling cross-platform algorithm validation, and (3) intact clinical workflow annotations including 19,705 technician-confirmed spike events and 2,757 seizure markers, enabling study of real-world annotation practices. We preserve these workflow-native annotations deliberately: while they lack the consistency of multi-expert research labels, they capture the noise, variability, and practical constraints under which automated systems must ultimately operate (Figure 1).

Table 2. Comparison with existing public EEG datasets.

Dataset	Patients	Sessions	Hours	Hardware	Recording types	Annotation style
CHB-MIT^3^	23	23	~982	Unknown	Inpatient	Expert seizure labels
Bonn^4^	5	5	~0.6	Intracranial	Research	Segment-level labels
Siena^5^	14	14	~128	Unknown	Inpatient	Expert seizure labels
TUH EEG Corpus^7^	~15,000	~25,000	~25,000	Natus NicoletOne	Primarily inpatient	Clinical reports
Neurotech (this work)	**1,744**	**8,410**	**77,575**	**Natus/Xltek**	**Routine + ambulatory + ICU**	**Workflow-native**

Methods

Patient population and clinical context

The dataset comprises all clinical EEG recordings performed at Neurotech between 2021 and 2025. No inclusion or exclusion criteria were applied; this cohort represents the full clinical caseload. Recording types span routine outpatient EEGs (typically <1 hour), ambulatory monitoring studies (1-24 hours), and prolonged continuous EEG monitoring in the intensive care unit (ICU; >24 hours). The 1,744 unique patients in the released dataset contributed 8,410 EEG recordings, with 76% of patients (1,320) having multiple recordings (median 4 per patient, interquartile range 2-9).

Recording hardware and protocol

All recordings were acquired using Natus/Xltek NeuroWorks EEG systems. Electrodes were placed according to the standard International 10-20 system, with typical montages including 25-29 channels: Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz, Pz, Cz, A1, A2, T1, T2, F11, F12, and an electrocardiogram (ECG) channel. Most files (5,979 of 8,410 with parseable headers; 71%) contain 29 channels including the EDF+ annotation signal. Signals were sampled at 256 Hz and stored in European Data Format (EDF+C, continuous).

Annotation methodology

EEG technicians placed annotations during routine clinical workflow using the Natus/Xltek annotation system. Three annotation types are present:

1. **Event markers** (`@Spike`, `@Seizure`): point-in-time markers for detected events, likely including both Persyst automated detections and technician-confirmed events.
2. **Technician clips** (`@Clip`): segments of interest selected by the technician for physician review, typically accompanied by descriptive labels (e.g., "Awake", "Tech Event Type 1: Generalized Sharp Waves").
3. **Free-text observations** (prefixed `NT-`): narrative clinical descriptions such as "NT-Bi-occipital S/W, right dominant" or "NT-Right occipital S/W."

Additional annotations document posterior dominant rhythm frequency and activation procedures (eyes open/closed, photic stimulation, hyperventilation). These are single-reader clinical workflow annotations, not multi-expert research labels. Inter-rater reliability was not assessed. While this limits their use as gold-standard evaluation labels, it enables study of real-world annotation practices including human-AI collaboration in EEG reading -- a research direction impossible with curated labels alone.

De-identification

We performed de-identification in compliance with HIPAA Safe Harbor standards, addressing three categories of protected health information (PHI):

Header scrubbing. We removed patient name, identifier, date of birth, case number, and technician identifiers from all EDF headers, replacing the local patient identification field with X X X X.

Date shifting. We shifted all recording dates by a random per-patient offset drawn uniformly from [-365, +365] days; times of day were preserved. The same offset was applied consistently across all recordings and annotation timestamps for a given patient.

Free-text scrubbing. We applied a two-tier name scrubber to annotation text: (1) each patient's own first and last name was matched and replaced with [NAME] regardless of word length, and (2) a broad dictionary of all first and last names in the dataset (4+ characters, excluding common medical terms) detected additional name occurrences. Dates embedded in annotation text were detected via pattern matching and shifted by the same per-patient offset. We verified de-identification by automated audit of all output files (see Technical Validation).

A linking table mapping de-identified identifiers to original identifiers is maintained securely and not published.

Clinical metadata extraction

Clinical documentation was available as scanned PDF packets for 1,629 of 1,741 patients (94%). Each packet contained a Neurotech technologist scan report, hourly monitoring logs, referring physician intake forms, and in many cases clinical progress notes from the referring neurologist. We developed a three-stage extraction pipeline: (1) text extraction from PDFs using pdftotext with OCR via Tesseract for scanned pages (80% of documents required OCR), (2) document segmentation into sub-document types using regex-based landmark detection, and (3) structured field extraction using a combination of deterministic regex parsers for standardized report sections and Gemini 2.5 Flash for narrative clinical text. The pipeline identified 12,440 sub-documents across 11 document types, extracting EEG findings (posterior dominant rhythm, epileptiform discharges, seizure descriptions), referral diagnoses (ICD-10 codes), patient demographics, medication lists, and hour-by-hour EEG monitoring data. All extraction code is available in the project repository.

Ethics

This project was conducted under IRB protocol number 2022P000417, with the BIDMC IRBs granting a waiver of consent, and under a BAA between BIDMC and Neurotech. The IRB approved the publication of the dataset in a de-identified form with access restricted by a data usage agreement prohibiting attempts at re-identification. The study also complied with the Declaration of Helsinki.

Data formatting

We converted the dataset to BIDS-EEG format (version 1.7.0)[8] and assigned each patient a de-identified identifier (sub-NeurotechN). Each EDF recording segment constitutes a separate session (ses-N), numbered sequentially per patient. For each session, the following files are provided (Figure 1):

- *_task-EEG_eeg.edf: de-identified EEG recording
- *_task-EEG_eeg.json: recording metadata (sampling frequency, channel counts, duration)
- *_task-EEG_channels.tsv: channel names, types, units, and filter settings
- *_task-EEG_Xltek.csv: technician annotations with shifted timestamps (when present)
- *_scans.tsv: session-level acquisition time

Dataset-level files include dataset_description.json, participants.tsv, participants.json, and a README. The full BIDS dataset comprises 80,382 files totaling 3.95 TB of storage.

Data Records

The dataset is hosted on the Brain Data Science Platform (BDSP) at s3://bdsp-opendata-repository/EEG/bids/Neurotech/. Table 1 summarizes the dataset characteristics.

Table 1. Patient and study characteristics. Clinical metadata was extracted from technician scan reports and referring physician documentation for the subset of patients with available clinical records (n=1,629 of 1,741 patients, 94%). Age and sex were available for 954 and 924 patients respectively. EEG findings are from technologist scan reports. EEG recording statistics are from the full BIDS dataset.

Characteristic	**Value**	**Notes**
Patients		
Unique patients	1,741	
With clinical documentation (EHR)	1,629	(94%)
Age at first EEG, median (IQR)	26.8 (13.2-48.2)	n=954
0-1 years	18	(2%)
2-5 years	58	(6%)
6-12 years	158	(17%)
13-17 years	126	(13%)
18-64 years	499	(52%)
65+ years	95	(10%)
Sex		n=924
Male	438	(47%)

Female	485	(52%)
Other	1	(0%)
Referral indications (ICD-10)		n=3,924
Epilepsy (G40.x)	2,105	(54%)
Convulsions (R56.x)	555	(14%)
Altered awareness (R40.x)	84	(2%)
Abnormal movements (R25.x)	191	(5%)
Syncope (R55)	56	(1%)
Other	933	(24%)
Comorbidities (from clinical notes)		n=1,188 patients
Seizures/Epilepsy	983	
Anxiety	217	
Hypertension	183	
Depression	173	
ADHD	105	
Asthma	102	
Hyperlipidemia	69	
Developmental Delay	61	
Anemia	44	
Hypothyroidism	41	
Anti-seizure medications		
Patients on >= 1 ASM	906	
Levetiracetam	555	
Lamotrigine	244	
Diazepam	186	
Lacosamide	168	
Clobazam	139	
Gabapentin	127	
Oxcarbazepine	102	
Clonazepam	81	
Topiramate	80	
Zonisamide	68	
EEG recordings		
Total recordings	8,040	
Total recording hours	74,321	
Recording duration, median (IQR)	3.3 (0.4-12.8)	hours
Recordings per patient, median (IQR)	3.0 (1.0-6.0)	
Patients with multiple recordings	1,270	(73%)
EEG findings (from tech reports)		n=2,710 studies
Normal	826	(30%)
Abnormal	1,286	(47%)
Interictal epileptiform discharges	1,418	studies
Focal/generalized slowing	1,135	studies
Electrographic seizures captured	779	studies

IQR = interquartile range; ASM = anti-seizure medication; EHR = electronic health record.

The majority of recordings (55%) fall in the 1-24 hour range consistent with ambulatory or short-term monitoring, while 34% are routine outpatient EEGs under one hour and 11% are prolonged continuous monitoring studies exceeding 24 hours (Supplementary Table 1, Figure 2A). In addition to the 8,410 recordings containing signal data, the BIDS dataset includes 10,397 header-only EDF files originating from the Natus/Xltek source export. These zero-record stubs are produced by the acquisition system at recording-session boundaries and aborted-start events; they contain a valid EDF header with channel definitions but no data records. We release them alongside the full-signal recordings to preserve session-level integrity for users wishing to reconstruct complete clinical visits, but they should be filtered out (e.g., on n_records > 0) for any signal-level analysis.

Technician clips and spike markers are the most common annotation types, followed by free-text clinical observations and activation procedure documentation (Supplementary Table 2, Supplementary Figure 1A).

Technical Validation

Data completeness and quality

All 8,410 EEG recordings were validated for EDF format compliance during BIDS conversion. Channel configuration is consistent across recordings, with all readable files containing 22-30 channels at 256 Hz (median: 29 channels). An additional 10,397 EDF files in the BIDS tree are zero-record header stubs from the Natus/Xltek source export (recording boundaries, interrupted sessions); they are documented in Data Records and should be filtered out for signal-level analyses. Signal quality metrics beyond format compliance (e.g., impedance values, artifact rates) are not reported; users should apply their own quality control procedures appropriate to their use case.

Annotation completeness

Of the 8,410 EEG recordings, 5,065 (60%) have at least one technician annotation file. Across the patient population, 1,742 of 1,744 subjects (99.9%) have at least one annotated recording. Annotation density varies widely: the median annotated recording contains 8 annotations (IQR 3-21), corresponding to 1.13 annotations per hour (IQR 0.47-2.78), with routine EEGs typically showing higher per-hour density and prolonged monitoring studies more total events (Figure 2C, Supplementary Figure 1A).

De-identification verification

We verified de-identification through automated audit of all output files:

4. **EDF headers**: We re-read all de-identified EDF files and confirmed that the local patient identification field contained only `X X X X`, with no residual patient names, identifiers, or unshifted dates (Supplementary Figure 2).
5. **Annotation text**: We scanned all 83,714 annotation text entries from the source data for potential PHI. Pattern matching identified annotations containing embedded dates (all shifted in output) and annotations containing patient first or last names (all replaced with `[NAME]` in output). No medical record numbers, phone numbers, or other identifiers were detected in the released data.
6. **File structure**: Output file and directory names contain only de-identified subject identifiers (`sub-NeurotechN`) and session numbers.

Supplementary Table 3 illustrates the de-identification process for a representative recording.

Usage Notes

Recommended use cases

This dataset supports three primary research directions. (1) **AI/ML development**: 83,714 technician annotations provide pre-existing labels for spike and seizure detection tasks (19,705 spike events, 2,757 seizure events), while the mix of recording types (routine, ambulatory, ICU) enables training algorithms that generalize across clinical settings. (2) **Clinical EEG research**: the unselected cohort enables epidemiological studies of EEG finding prevalence and studies of annotation variability in clinical workflow. (3) **Methodological development**: automated sleep staging can be applied post-hoc to the multi-day recordings, enabling large-scale study of interictal epileptiform discharge (IED) rates across sleep-wake states.

Limitations

The dataset has several important limitations. First, it originates from a **single center** with one hardware platform; generalization to other institutions requires caution. Second, annotations are **clinical workflow annotations** placed by technicians during routine practice, not multi-expert research labels; they should not be treated as gold-standard labels for algorithm evaluation without independent validation. This is both a strength and a limitation: while it enables study of real-world labeling practices, it means labels cannot be treated as ground truth. Third, clinical metadata (demographics, diagnoses, comorbidities, medications, and EEG interpretation reports) was extracted from scanned clinical documentation using a combination of OCR, regex parsing, and large language model (LLM) extraction. Coverage varies by field: age is available for 55% of patients, sex for 53%, referral diagnosis codes for 66%, and anti-seizure medication data for 52%. Fields extracted from handwritten forms have lower accuracy than those from typed clinical notes. Fourth, **annotation completeness is heterogeneous**: 60% of EEG recordings (5,039 of 8,410) have associated annotation files, with 99.9% of patients (1,742 of 1,744) having at least one annotated recording. Fifth, the BIDS tree includes **10,397 zero-record header stubs** alongside the 8,410 EEG recordings (Natus/Xltek source artifact); users should filter on n_records > 0 or file size when constructing analysis cohorts.

Future directions

Future versions will expand clinical metadata coverage to the full patient cohort (the current release includes clinical data for patients with last names A-H; the remaining patients are pending data transfer), add multi-expert re-annotation of a validation subset to establish inter-rater reliability, apply automated sleep staging to multi-day recordings, and develop a curated bank of teaching cases with an interactive web application for EEG education.

Data access

The dataset is available through BDSP at s3://bdsp-opendata-repository/EEG/bids/Neurotech/. Access requires a data use agreement (DUA); registration details are available at https://bdsp.io. The dataset is formatted according to BIDS-EEG v1.7.0 and can be read using standard tools including MNE-Python, MNE-BIDS, pyedflib, and edfio.

Code availability

Code for the BIDS conversion pipeline, de-identification procedures, and annotation extraction is available at https://github.com/bdsp-core/Neurotech-EEG-Wrangling.

Figure Legends

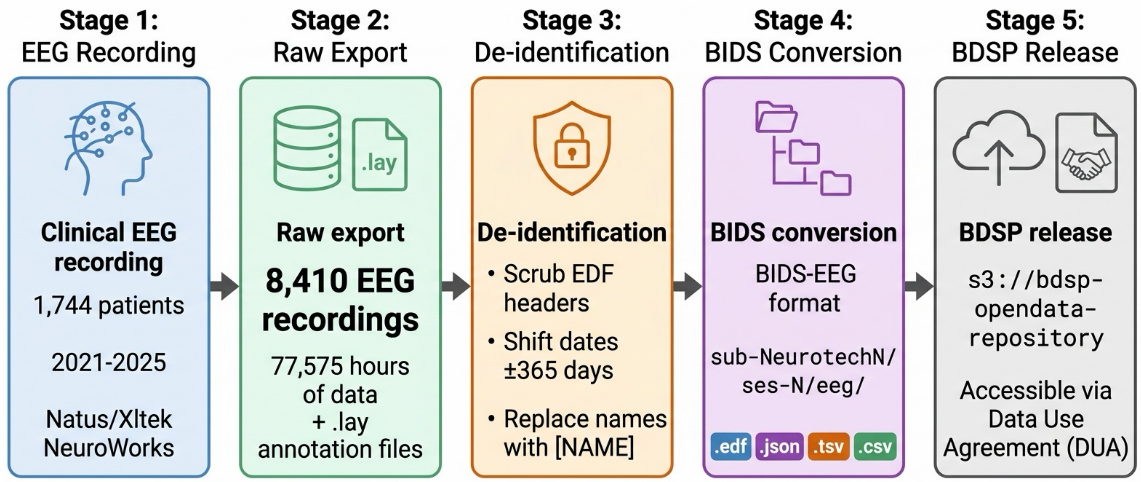

Figure 1. Data pipeline from clinical recording to public release. 8,410 EEG recordings from 1,744 patients (2021-2025) totaling 77,575 hours were de-identified through header scrubbing, per-patient date shifting (uniform random offset of +/- 365 days), and automated name replacement in free text, converted to BIDS-EEG format, and released through the Brain Data Science Platform (BDSP) via data use agreement.

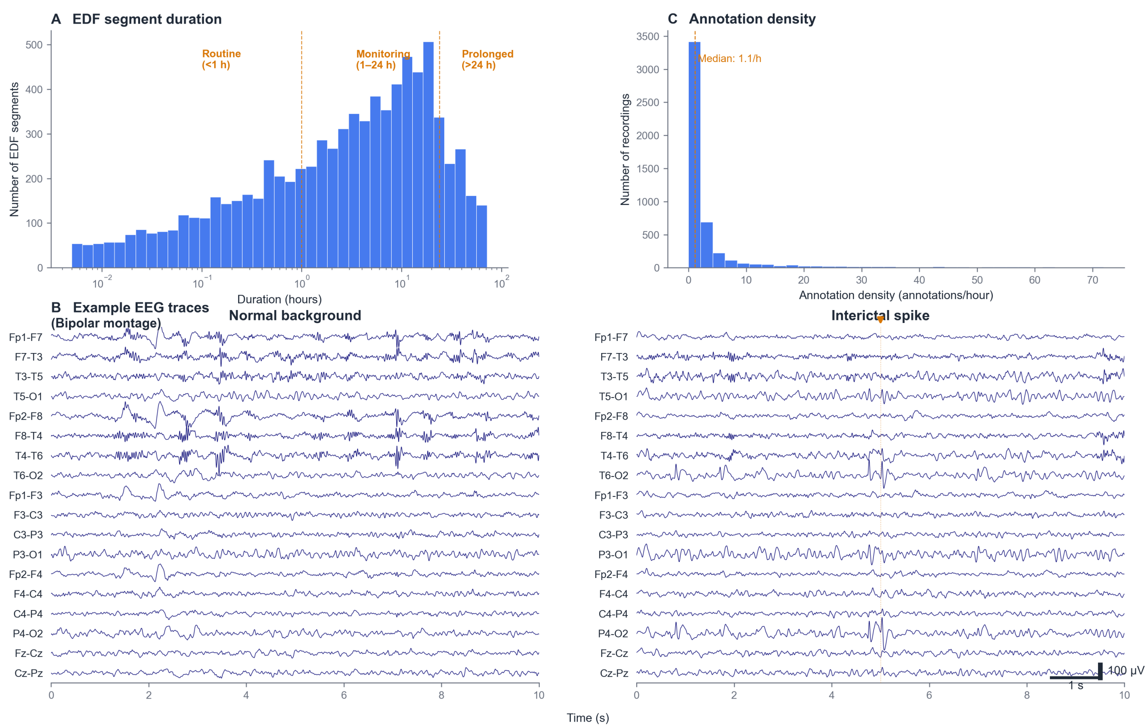

Figure 2. Dataset characteristics. (A) Distribution of EDF segment duration on a logarithmic scale (n = 8,410 EDFs with parseable signal data). Dashed lines indicate approximate boundaries between routine EEGs (<1 hour), ambulatory/short-term monitoring (1-24 hours), and prolonged continuous monitoring (>24 hours). (B) Example EEG traces from a representative recording in standard bipolar montage

showing normal background activity (left) and an interictal spike with technician annotation (right). Eight channels from the standard 10-20 montage are displayed; scale bars indicate 100 microvolts and 1 second. The annotation "NT-Bi-occipital S/W, right dominant" illustrates the free-text clinical observations preserved in the dataset. (C) Distribution of annotation density (annotations per hour) across 5,000 sessions with both technician annotations and known duration; dashed line indicates the median (1.13 per hour).

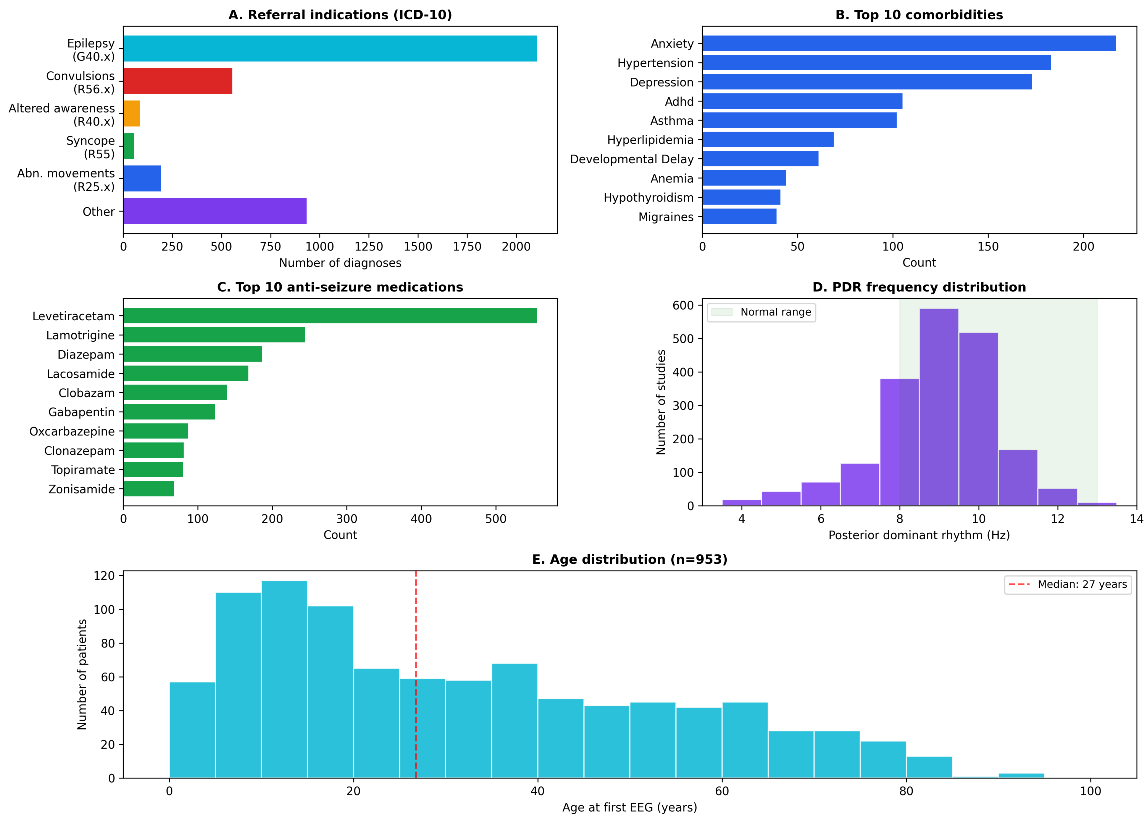

Figure 3. Clinical characteristics of the Neurotech EEG Dataset. (A) Distribution of referral indications by ICD-10 code category (n=3,924 diagnosis codes from intake forms). Epilepsy (G40.x) is the most common indication, followed by unspecified convulsions (R56.x). (B) Ten most frequent comorbidities extracted from clinical progress notes, excluding seizure-related diagnoses. (C) Ten most frequently prescribed anti-seizure medications. Levetiracetam is the most common, consistent with its status as a first-line agent. (D) Distribution of posterior dominant rhythm (PDR) frequency extracted from technologist scan reports (n=2,043 studies with extractable PDR). The peak at 9-10 Hz and right-skewed distribution are consistent with normal physiological values; the green shading indicates the normal adult range (8-13 Hz). (E) Age distribution at first EEG (n=953 patients with available date of birth), showing a bimodal pattern with a pediatric peak (5-15 years) and a broad adult distribution. Median age 27 years (dashed line).

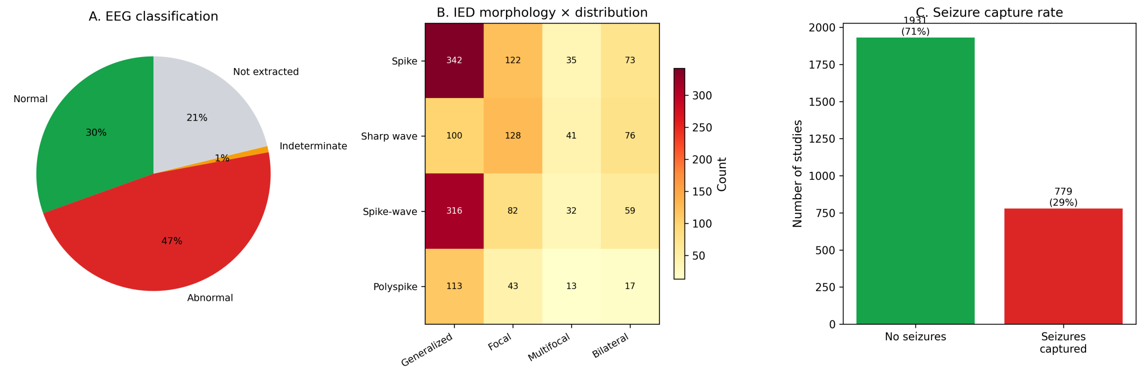

Figure 4. EEG findings summary. (A) EEG classification by technologist impression: 47% of studies were classified as abnormal, 30% as normal, and 21% were not classifiable from the extracted text. (B) Heatmap of interictal epileptiform discharge (IED) morphology by spatial distribution across 1,418 studies with documented IEDs. Values represent the number of studies with each morphology-distribution combination. (C) Seizure capture rate: 779 of 2,710 studies (29%) had electrographic seizures documented by the scanning technologist.

References

7. Noachtar, S. & Remi, J. The role of EEG in epilepsy: a critical review. *Epilepsy Behav.* 15, 22-33 (2009).
8. Westover, M.B. et al. Machine learning for electroencephalography: current status and future directions. *J. Clin. Neurophysiol.* (2023).
9. Shoeb, A. & Guttag, J. Application of machine learning to epileptic seizure detection. *Proc. ICML* (2010). [CHB-MIT]
10. Andrzejak, R.G. et al. Indications of nonlinear deterministic and finite-dimensional structures in time series of brain electrical activity. *Phys. Rev. E* 64, 061907 (2001). [Bonn dataset]
11. Detti, P. et al. EEG synchronization analysis for seizure prediction: a study on data of noninvasive recordings. *Processes* 8, 846 (2020). [Siena dataset]
12. Gemein, L.A.W. et al. Machine-learning-based diagnostics of EEG pathology. *NeuroImage* 220, 117021 (2020).
13. Obeid, I. & Picone, J. The Temple University Hospital EEG Data Corpus. *Front. Neurosci.* 10, 196 (2016).
14. Pernet, C.R. et al. EEG-BIDS, an extension to the brain imaging data structure for electroencephalography. *Sci. Data* 6, 103 (2019).

Conflicts of Interest

MBW is a co-founder, scientific advisor, consultant to, and has personal equity interest in Beacon Biosignals. KM, CP, and MG are employees of Neurotech. The other authors declare that they have no conflicts of interest.

Funding

MBW receives research funding from the NIH (RF1AG064312, RF1NS120947, R01AG073410, R01HL161253, R01NS126282, R01AG073598, R01NS131347, R01NS130119, R01NS131347). DMG receives research funding from the NIH (K23NS124656).

Word count: ~3,800 (excluding tables, references, abstract, figure legends, and supplementary material)

Supplementary Material

Supplementary Tables

Supplementary Table 1. Recording duration categories. The majority of recordings represent ambulatory or short-term monitoring studies.

Category	Duration	N segments (%)	Likely clinical setting
Routine	< 1 hour	2,842 (34%)	Outpatient EEG
Short monitoring	1 - 24 hours	4,652 (55%)	Ambulatory or short-term
Prolonged monitoring	> 24 hours	916 (11%)	Inpatient continuous EEG or multi-day ambulatory

Percentages computed over 8,410 EDF files containing recoverable signal data.

Supplementary Table 2. Annotation categories. Technician clips and spike markers are the most frequent annotation categories.

Category	Events	Description
Technician clips	19,902	Segments selected for physician review
Spike markers	19,705	Interictal epileptiform discharge detections
Neurotech free-text comments	8,932	Free-text EEG finding descriptions
Activation procedures	8,149	Eyes open/closed, photic, hyperventilation
Sharp waves	8,027	Sharp wave or sharp-slow-wave complexes
Slowing	7,066	Focal or diffuse slowing
Generalized patterns	4,567	Generalized discharges
Spike-wave	3,978	Spike-wave complexes
Seizure markers	2,757	Electrographic seizure events
Posterior dominant rhythm	1,892	PDR frequency notations
Artifact	1,585	Technical or physiological artifacts
Burst-suppression	1,535	Burst-suppression (ICU recordings)
Focal	904	Focal patterns
Normal	201	"Normal" notations
Epileptiform	88	Other epileptiform
Periodic	68	Periodic discharges (LPDs/GPDs)

Annotation text was parsed into categories using keyword matching. Categories are not mutually exclusive; a single annotation may contribute to multiple categories. Total annotation events: 83,714 across 5,065 sessions with annotation files.

Supplementary Table 3. De-identification example. Header field transformation and annotation scrubbing for a representative recording.

Field	Original	De-identified
Patient identification	12-34567 X 15-MAR-2023 Doe_Jane	X X X X
Recording identification	Startdate 15-MAR-2023 Record_stopped...	Startdate 22-JUN-2023 X X X
Start date	15.03.23	22.06.23 (shifted +99 days)
Start time	14.32.08	14.32.08 (preserved)
Annotation text	Patient event Jane had a blank stare	Patient event [NAME] had a blank stare
Annotation timestamp	2023-03-15T14:33:45	2023-06-22T14:33:45 (shifted)

Names and dates shown are fictitious examples. Actual date shifts are random per patient (uniform integer in [-365, +365] days) and applied consistently across all files for that patient.

Supplementary Table 4. Detailed EEG findings. Breakdown of posterior dominant rhythm, interictal epileptiform discharges, slowing, seizures, and patient-reported events extracted from technologist scan reports (n=2,710 studies with extractable reports).

Finding	Value	Notes
Posterior dominant rhythm		
PDR extractable	2,043	of 2,710 studies
PDR frequency, median (IQR)	9.0 (8.0-10.0)	Hz
Normal range (8-13 Hz)	1,713	(84%)
Slow (<8 Hz)	303	(15%)
Interictal epileptiform discharges	1,418	of 2,710 studies (52%)
Morphology		
Spike	602	
Sharp wave	521	
Spike-and-wave	508	
Polyspike	158	
Distribution		
Generalized	420	
Focal	274	
Multifocal	85	
Bilateral independent	29	
Laterality		
Left	477	
Right	421	
Bilateral	164	
Region		
Temporal	537	
Frontal	308	
Central	183	
Parietal	72	
Occipital	63	
Abnormal slowing	1,135	of 2,710 studies (42%)
Electrographic seizures	779	of 2,710 studies (29%)
Patient-reported events	1,416	
With timestamp	1,378	(97%)

Morphology, distribution, laterality, and region were extracted from free-text descriptions using regex pattern matching. Categories are not mutually exclusive; a single description may contain multiple morphology or distribution terms.

Supplementary Table 5. Hourly EEG monitoring data. Summary of technician monitoring activity extracted from TrackIT monitoring logs in technologist scan reports.

Metric	**Value**
Studies with monitoring data	2,192
Total monitoring hours documented	102,185
Hours with recording active	94,369 (92%)
Hours with recording gap	7,816 (8%)
Monitoring events	59,664
EEG reviewed notes	46,309
General notes	13,164
Equipment failures	191
Reviewers per study, median (IQR)	4.0 (2.0-7.0)

Each monitoring hour was documented individually by the assigned EEG technician, including impedance readings, recording status, battery levels, and free-text observations. Hour-by-hour data enables reconstruction of the complete monitoring timeline for each study.

Supplementary Figure Legends

Supplementary Figure 1. Annotation and recording summary. (A) Frequency of 83,714 annotation categories extracted from technician annotation files by keyword matching (see Supplementary Table 2). Categories are not mutually exclusive; a single annotation may contribute to multiple categories. (B) Comparison of the Neurotech EEG Dataset with existing public EEG datasets by number of patients (blue) and total recording hours (orange) on a logarithmic scale. The Neurotech dataset provides the largest total recording hours of any single-center public clinical EEG corpus.

Supplementary Figure 2. De-identification of EDF header fields. Each row shows a header field before (containing protected health information) and after de-identification. Patient names are replaced with placeholders, identifiers are reassigned, dates are shifted by a consistent per-patient random offset, and technician identifiers are removed. Equipment information is preserved. All examples shown are fictitious.

Supplementary Figure 3. Hourly EEG monitoring characteristics. (A) Distribution of total monitoring duration across 2,192 studies with hour-by-hour monitoring data. Peaks at 24 and 48 hours reflect standard ordered monitoring durations. (B) Recording activity by hour of day across all 102,185 documented monitoring hours. Light shading shows total hours logged; dark shading shows hours with active recording. The slight reduction in active recording during daytime hours reflects battery replacements, electrode troubleshooting, and brief patient disconnections documented in the monitoring logs.